

Лабораторная работа №6

Имитационное моделирование

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Целью данной лабораторной работы является реализация модели эпидемии SIR с использованием сетей Петри и анализ результата действия данной модели.

Предварительно проверим правильность структуры нашего проекта (рис. 1).

```
(nikita@nikita) ~/labs/lab06/project/scripts
$ julia test_setup.jl
Activating project at `~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project`
✓ Проект активирован: /home/nikita/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project

Проверка пакетов:
✓ DrWatson
✓ DifferentialEquations
✓ Plots
✓ DataFrames
✓ CSV
✓ JLO2
✓ Literate
✓ IJulia
✓ BenchmarkTools
✓ Quarto

Структура проекта:
Корень: /home/nikita/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project
Данные: /home/nikita/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/data
Скрипты: /home/nikita/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/src
Графики: /home/nikita/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/plots
```

→ Infection: $S + I \rightarrow I + I$ (экспонента β);
→ recovery: $I \rightarrow R$ (экспонента γ).

Рисунок 1: Проверка структуры проекта

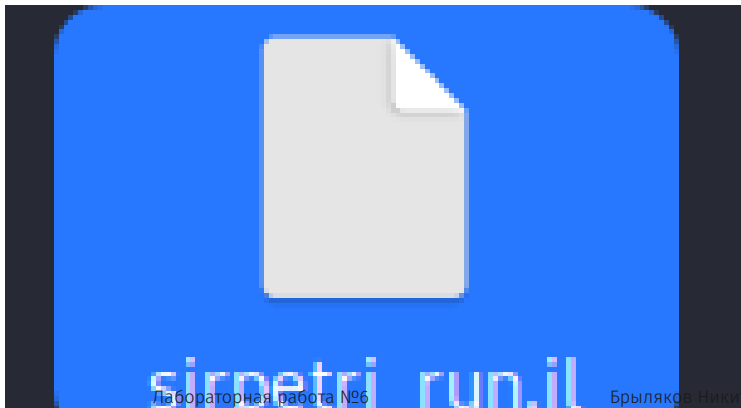
Создадим файл `src/SIRPetri.jl` с реализацией вычислительной логики модели (рис. 2).

A terminal window with a dark background. The prompt is `(nikita@nikita)-[~/.../labs/lab06/project/src]`. The command `$ touch SIRPetri.jl` has been entered and executed. The output `lab02` is visible in the background.

```
(nikita@nikita)-[~/.../labs/lab06/project/src]  
$ touch SIRPetri.jl  
lab02
```

Рисунок 2: Код модели

Создадим файл `scripts/sirpetri_run.jl`. Скрипт выполняет один базовый эксперимент с фиксированными параметрами $\beta = 0.3$, $\gamma = 0.1$ и запускает два типа симуляции: детерминированную (решение ОДУ) — даёт плавную усреднённую динамику и стохастическую (алгоритм Гиллеспи) — учитывает случайные флуктуации (рис. 3).

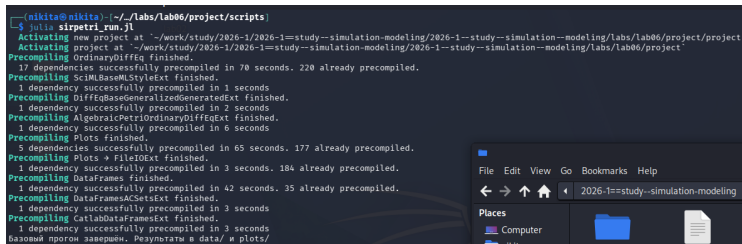


Запустим скрипт (рис. 4).

```
nikita@nikita:~/./labs/lab06/project/scripts$ julia tangle.jl sirpetri_run.jl
Activating project at "/work/study/2026-1/study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project"
Generating esi: sirpetri_run.jl
Info: generating plain script file from "/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/scripts/sirpetri_run.jl"
Info: writing result to "/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/scripts/sirpetri_run.jl"
v Часовой пояс: /home/nikita/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/scripts/sirpetri_run.jl
Info: generating markdown page from "/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/scripts/sirpetri_run.jl"
Info: writing result to "/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/markdown/sirpetri_run/sirpetri_run.qmd"
v Quarto: /home/nikita/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/markdown/sirpetri_run/sirpetri_run.qmd
Info: generating notebook from "/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/scripts/sirpetri_run.jl"
Info: writing result to "/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/notebooks/sirpetri_run/sirpetri_run.ipynb"
v Notebook: /home/nikita/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/notebooks/sirpetri_run/sirpetri_run.ipynb
Готово! Все файлы созданы.
```

Рисунок 4: Базовый прогон модели

Создадим производные форматы с помощью скрипта `tangle.jl` (рис. 5).



```
(nikita@nikita)-[~/./labs/lab06/project/scripts]
$ julia sirpetri_run.jl
Activating new project at `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/project`
Activating project at `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project`
Precompiling OrdinaryDiffEq finished.
17 dependencies successfully precompiled in 70 seconds. 220 already precompiled.
Precompiling SciMLBaseMLStyleExt finished.
1 dependency successfully precompiled in 1 seconds
Precompiling DiffEqBaseGeneralizedGeneratedExt finished.
1 dependency successfully precompiled in 2 seconds
Precompiling AlgebraicPetriOrdinaryDiffEqExt finished.
1 dependency successfully precompiled in 6 seconds
Precompiling Plots finished.
5 dependencies successfully precompiled in 65 seconds. 177 already precompiled.
Precompiling Plots → FileIOExt finished.
1 dependency successfully precompiled in 3 seconds. 184 already precompiled.
Precompiling DataFrames finished.
1 dependency successfully precompiled in 42 seconds. 35 already precompiled.
Precompiling DataFramesACSetsExt finished.
1 dependency successfully precompiled in 3 seconds
Precompiling CatlabDataFramesExt finished.
1 dependency successfully precompiled in 3 seconds
Базовый прогон завершен. Результаты в data/ и plots/
```

Рисунок 5: Создание производных форматов

Запустим файл `irynb` в jupyter-notebook (рис. 6).

```
In [*]: using Pkg
        Pkg.activate("../project")
        using DrWatson
        @quickactivate "project"
        using Random
        include(scrdir("SIRPetri.jl"))
        using .SIRPetri
        using DataFrames, CSV, Plots
```

Параметры

```
In [ ]: β = 0.3
        γ = 0.1
        tmax = 100.0
```

Создаём сеть

```
In [ ]: net, u0, states = SIRPetri.build_sir_network(β, γ)
```

Детерминированная симуляция

```
In [ ]: df_det = SIRPetri.simulate_deterministic(net, u0, (0.0, tmax), saveat = 0.5, rates = [β, γ])
        CSV.write(datadir("sir_det.csv"), df_det)
```

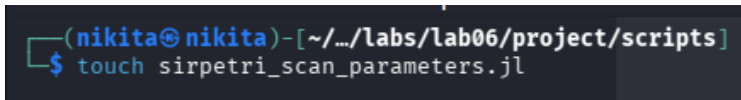
Стохастическая симуляция

```
In [ ]: Random.seed!(123)
        df_stoch = SIRPetri.simulate_stochastic(net, u0, (0.0, tmax), rates = [β, γ])
        CSV.write(datadir("sir_stoch.csv"), df_stoch)
```

Рисунок 6: Запуск файла в jupyter-notebook

Коэффициент заражения β

Создадим файл `scripts/sirpetri_scan_parameters.jl`. Скрипт исследует чувствительность модели к изменению параметра β (скорости заражения). Для каждого значения β из диапазона `0.1:0.05:0.8` запускается детерминированная симуляция ($\gamma = 0.1$ фиксирован). Для каждого прогона вычисляются: максимальное число инфицированных (пик эпидемии) – `peak_I` и конечное число выздоровевших – `final_R` (рис. 7).



```
(nikita@nikita)-[~/.../labs/lab06/project/scripts]  
$ touch sirpetri_scan_parameters.jl
```

Рисунок 7: Коэффициент заражения β

Запустим скрипт (рис. 8).

A terminal window with a dark background. The prompt is '(nikita@nikita)-[~/./labs/lab06/project/scripts]'. The user enters '\$ julia sirpetri_scan_parameters.jl'. The output shows 'Activating new project at ~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/project' and 'Activating project at ~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project'. At the bottom, it says 'Сканирование β завершено. Результат в data/sir_scan.csv'.

```
(nikita@nikita)-[~/./labs/lab06/project/scripts]
$ julia sirpetri_scan_parameters.jl
Activating new project at "~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/project"
Activating project at "~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project"
Сканирование  $\beta$  завершено. Результат в data/sir_scan.csv
```

Рисунок 8: Коэффициент заражения β

Создадим производные форматы с помощью скрипта `tangle.jl` (рис. 9).

```
(nikita@nikita) [~/labs/lab06/project/scripts]
$ julia tangle.jl sirpetri_scan_parameters.jl
Activating project at "~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project"
Resolving dependencies...
[ Info: generating plain script file from "~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/scripts/sirpetri_scan_parameters.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/scripts/sirpetri_scan_parameters.jl"
[ Info: generating markdown page from "~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/scripts/sirpetri_scan_parameters.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/markdown/sirpetri_scan_parameters/sirpetri_scan_parameters.jl"
[ Info: generating notebook from "~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/scripts/sirpetri_scan_parameters.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/notebooks/sirpetri_scan_parameters/sirpetri_scan_parameters.jl"
```

Рисунок 9: Создание производных форматов

Запустим файл `irunb` в `jupyter-notebook` (рис. 10).

```
In [1]: using Pkg
        Pkg.activate("../project")
        using DrWatson
        @quickactivate "project"
        include(scrdir("SIRPetri.jl"))
        using .SIRPetri
        using DataFrames, CSV, Plots

        Activating project at `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation--modeling/
        labs/lab06/project/notebooks/project`
        Activating project at `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation--modeling/
        labs/lab06/project/notebooks`

The activated project did not match asserted name. Current project name is nothing while the asserted name is projec
t.

Stacktrace:
 [1] error{s::String)
    @ Base ./error.jl:44
 [2] quickactivate(path::String, name::String)
    @ DrWatson ~/.julia/packages/DrWatson/20F5p/src/project_setup.jl:138
 [3] top-level scope
    @ In[1]:4
 [4] eval(m::Module, e::Any)
    @ Core ./boot.jl:489

Диапазон  $\beta$ 

In [2]:  $\beta$ _range = 0.1:0.05:0.8
         $\gamma$ _fixed = 0.1
        tmax = 100.0

        results = []
        for  $\beta$  in  $\beta$ _range
            net, u0, _ = SIRPetri.build_sir_network( $\beta$ ,  $\gamma$ _fixed)
            df = SIRPetri.simulate_deterministic(net, u0, (0.0, tmax), saveat = 0.5, rates = [ $\beta$ ,  $\gamma$ _fixed])
            peak_I = maximum(df.I)
```

Рисунок 10: Запуск файла в `jupyter-notebook`

Создадим файл `scripts/sirpetri_animate.jl`. Скрипт создаёт GIF-анимацию, показывающую, как со временем меняется количество людей в каждой из трёх групп (S,I,R). Анимация позволяет наглядно увидеть распространение эпидемии, пик и спад (рис. 11).

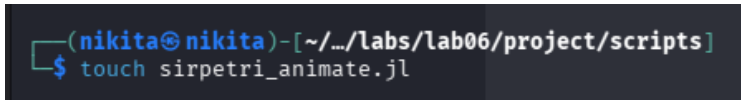


Рисунок 11: Анимация детерминированной динамики

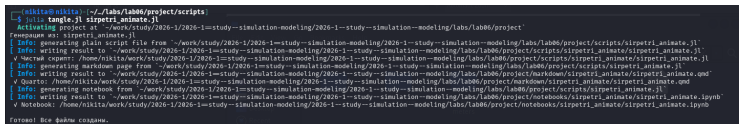
Запустим скрипт (рис. 12).



```
(nikita@nikita)~/.julia/labs/lab06/project/scripts]
$ julia sirpetri_animate.jl
Activating project at "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project"
Сканирование β завершено. Результат в data/sir_scan.csv
```

Рисунок 12: Анимация детерминированной динамики

Создадим производные форматы с помощью скрипта `tangle.jl` (рис. 13).



```
(nikita@nikita) ~/work/study/2026-1/2026-1-study-simulation-modeling/2026-1-study-simulation-modeling/labs/lab06/project
$ julia tangle.jl sirpetri_animate.jl
Activating project at "/work/study/2026-1/2026-1-study-simulation-modeling/2026-1-study-simulation-modeling/labs/lab06/project"
Генерация из: sirpetri_animate.jl
[ Info: generating plain script file from "/work/study/2026-1/2026-1-study-simulation-modeling/2026-1-study-simulation-modeling/labs/lab06/project/scripts/sirpetri_animate.jl"
[ Info: writing result to "/work/study/2026-1/2026-1-study-simulation-modeling/2026-1-study-simulation-modeling/labs/lab06/project/scripts/sirpetri_animate/sirpetri_animate.jl"
V Чистый скрипт: /home/nikita/work/study/2026-1/2026-1-study-simulation-modeling/2026-1-study-simulation-modeling/labs/lab06/project/scripts/sirpetri_animate/sirpetri_animate.jl
[ Info: generating markdown page from "/work/study/2026-1/2026-1-study-simulation-modeling/2026-1-study-simulation-modeling/labs/lab06/project/scripts/sirpetri_animate.jl"
[ Info: writing result to "/work/study/2026-1/2026-1-study-simulation-modeling/2026-1-study-simulation-modeling/labs/lab06/project/markdown/sirpetri_animate/sirpetri_animate.qmd"
V Quarto: /home/nikita/work/study/2026-1/2026-1-study-simulation-modeling/2026-1-study-simulation-modeling/labs/lab06/project/markdown/sirpetri_animate/sirpetri_animate.qmd
[ Info: generating notebook from "/work/study/2026-1/2026-1-study-simulation-modeling/2026-1-study-simulation-modeling/labs/lab06/project/scripts/sirpetri_animate.jl"
[ Info: writing result to "/work/study/2026-1/2026-1-study-simulation-modeling/2026-1-study-simulation-modeling/labs/lab06/project/notebooks/sirpetri_animate/sirpetri_animate.ipynb"
V Notebook: /home/nikita/work/study/2026-1/2026-1-study-simulation-modeling/2026-1-study-simulation-modeling/labs/lab06/project/notebooks/sirpetri_animate/sirpetri_animate.ipynb
Готово! Все файлы созданы.
```

Рисунок 13: Создание производных форматов

Анимация детерминированной динамики

Запустим файл `ipy.nb` в `jupyter-notebook` (рис. 14).

```
In [1]: using DrWatson
@quickactivate "project"
include(scrdir("SIRPetri.jl"))
using .SIRPetri
using DataFrames, CSV, Plots
```

Параметры (можно оставить ваши или настроить под конкретный запуск)

```
In [2]: β = 0.4
γ = 0.1
tmax = 50.0
```

```
Out[2]: 50.0
```

1. Запускаем симуляцию

```
In [3]: net, u0, _ = SIRPetri.build_sir_network(β, γ)
df = SIRPetri.simulate_deterministic(net, u0, (0.0, tmax), saveat = 0.5, rates = [β, γ])
```

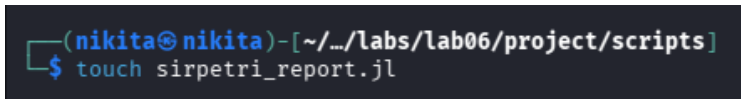
```
Out[3]: 101×4 DataFrame
```

76 rows omitted

Row	time	S	I	R
	Float64	Float64	Float64	Float64
1	0.0	990.0	10.0	0.0
2	0.5	4.71208e-7	952.326	47.6744
3	1.0	4.84332e-7	905.88	94.1199
4	1.5	-3.35068e-7	861.7	138.3
5	2.0	-2.01894e-7	819.674	180.326
6	2.5	3.7506e-7	779.698	220.302
7	3.0	-9.70587e-8	741.672	258.328

Рисунок 14: Запуск файла в `jupyter-notebook`

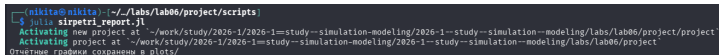
Создадим файл `scripts/sirpetri_report.jl`. Скрипт загружает ранее сохранённые результаты (`sir_det.csv`, `sir_stoch.csv`, `sir_scan.csv`) и строит сравнительные графики для итогового отчёта (рис. 15).

A terminal window with a dark background. The prompt is `(nikita@nikita)-[~/.../labs/lab06/project/scripts]`. The command entered is `$ touch sirpetri_report.jl`.

```
(nikita@nikita)-[~/.../labs/lab06/project/scripts]  
$ touch sirpetri_report.jl
```

Рисунок 15: Итоговый отчёт

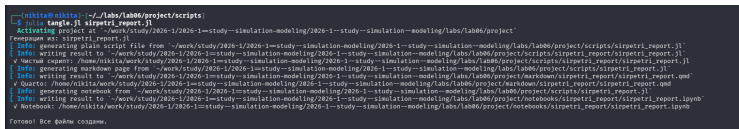
Запустим скрипт (рис. 16).

A terminal window with a dark background. The prompt is '(nikita@nikita) ~/labs/lab06/project/scripts |'. The command entered is '\$ julia sirpetri_report.jl'. The output shows 'Activating new project at `~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/project`' and 'Activating project at `~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project`'. At the bottom, it says 'Отчётные графики сохранены в plots/'.

```
(nikita@nikita) ~/labs/lab06/project/scripts |  
$ julia sirpetri_report.jl  
Activating new project at `~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/project`  
Activating project at `~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project`  
Отчётные графики сохранены в plots/
```

Рисунок 16: Итоговый отчёт

Создадим производные форматы с помощью скрипта `tangle.jl` (рис. 17).



```
(nikita@nikita): ~/./labs/lab06/project/scripts
$ julia tangle.jl sirpetri_report.jl
Activating project at "~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project"
resepapa@ns: sirpetri_report.jl
Info: generating plain script file from "~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/scripts/sirpetri_report.jl"
Info: writing result to "~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/scripts/sirpetri_report/sirpetri_report.jl"
✓ Числм скрипт: /home/nikita/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/scripts/sirpetri_report/sirpetri_report.jl
Info: generating markdown page from "~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/scripts/sirpetri_report.jl"
Info: writing result to "~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/markdown/sirpetri_report/sirpetri_report.qmd"
✓ Quarto: /home/nikita/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/markdown/sirpetri_report/sirpetri_report.qmd
Info: generating notebook from "~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/scripts/sirpetri_report.jl"
Info: writing result to "~/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/notebooks/sirpetri_report/sirpetri_report.ipynb"
✓ Notebook: /home/nikita/work/study/2026-1/2026-1=study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/notebooks/sirpetri_report/sirpetri_report.ipynb
Готово! Все файлы созданы.
```

Рисунок 17: Создание производных форматов

Запустим файл ipynb в jupyter-notebook (рис. 18).

```
In [1]: using Pkg
        Pkg.activate("../project")
        using DrWatson
        @quickactivate "project"
        using DataFrames, CSV, Plots

df_det = CSV.read(datadir("sir_det.csv"), DataFrame)
df_stoch = CSV.read(datadir("sir_stoch.csv"), DataFrame)
df_scan = CSV.read(datadir("sir_scan.csv"), DataFrame)

Activating new project at `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project/notebooks/project`
Activating project at `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation--modeling/labs/lab06/project`

Out[1]: 15x3 DataFrame
        Row  β      peak_I  final_R
        Float64 Float64  Float64
1      0.1  955.626  999.954
2      0.15 954.157  999.954
3      0.2  953.424  999.954
4      0.25 952.984  999.955
5      0.3  952.691  999.955
6      0.35 952.482  999.955
7      0.4  952.326  999.955
8      0.45 952.204  999.955
9      0.5  952.106  999.955
10     0.55 952.026  999.955
11     0.6  951.96  999.955
```

Рисунок 18: Запуск файла в jupyter-notebook

В ходе данной лабораторной работы мной была реализована модель эпидемии SIR с использованием сетей Петри и проанализирован результат действия данной модели.